PPM Dies Monitoring System

Technical Documentation

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Author: Development Team

Table of Contents

1. System Overview
2. System Flow Diagram
3. Database Schema
4. Core Logic
5. Alert System
6. Lot Progress Visualization
7. User Roles & Permissions
8. API Endpoints

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1. System Overview

1.1 Purpose

PPM Dies Monitoring adalah sistem pengendalian untuk:

- **Membuat PPM berdasarkan produksi aktual** mengacu pada standard stroke yang telah ditentukan
- **Melindungi perawatan cetakan (dies)** agar tidak melebihi batas stroke
- **Mengurangi part bermasalah** akibat kondisi dies yang tidak terpelihara

1.2 Technology Stack

Component	Technology
Backend	Laravel 11 (PHP 8.2+)
Frontend	React + Inertia.js
Database	MySQL/MariaDB
Styling	Tailwind CSS
Notifications	Laravel Notifications (Mail)

1.3 Key Features

- Real-time stroke accumulation tracking
- Automated PPM scheduling based on production
- Multi-level alert notifications (Orange & Red)

- Visual lot progress indicator
- Production log management (LHP)
- Role-based access control

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2. System Flow Diagram

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          PPM DIES MONITORING FLOW                        │
└──────────────────────────────────────────────────────────────────────────┘

                              ┌──────────────┐
                              │     PE       │
                              │(Production)  │
                              │Engineering)  │
                              └──────────────┘
                                     │
                                     ▼
                          ┌──────────────────────┐
                          │     MTN DIES         │
                          │  (Maintenance)       │
                          │  Input Data Dies     │
                          └──────────────────────┘
                                     │
                       ┌─────────────┼─────────────┐
                       │             │             │
                       ▼             ▼             ▼
              ┌──────────────┐ ┌──────────────┐ ┌──────────────┐
              │   Standard   │ │   Lot Size   │ │ Production   │
              │   Stroke     │ │   (PPM)      │ │    Logs      │
              │  (10,000)    │ │   (2,500)    │ │   (LHP)      │
              └──────────────┘ └──────────────┘ └──────────────┘
                       │             │             │
                       └─────────────┼─────────────┘
                                     │
                                     ▼
                       ┌──────────────────────────┐
                       │       SYSTEM CALCULATES  │
                       │  ────────────────────────│
                       │  Accumulation Stroke vs  │
                       │  Standard Stroke         │
                       │                          │
                       │  Total Lots = Standard / │
                       │               Lot Size   │
                       │  (10,000 / 2,500 = 4 lots)│
                       └──────────────────────────┘
                                     │
                       ┌─────────────┼─────────────┐
                       │                           │
                       ▼                           ▼
              ┌──────────────────┐          ┌──────────────────┐
              │   ORANGE ZONE    │          │     RED ZONE     │
```

```
        │   (Warning)        │          │    (Critical)      │
        │                    │          │                    │
        │  Stroke ≥          │          │  Stroke ≥          │
        │  Standard - 1 Lot  │          │  Standard Stroke   │
        │  (≥ 7,500)         │          │  (≥ 10,000)        │
        └────────────────────┘          └────────────────────┘
                  │                               │
                  ▼                               ▼
        ┌────────────────────┐          ┌────────────────────┐
        │  NOTIFY:           │          │  NOTIFY:           │
        │  • MGR/GM          │          │  • MGR/GM          │
        │  • MD              │          │  • MD              │
        │                    │          │  • MTN Dies        │
        └────────────────────┘          └────────────────────┘
                  │                               │
                  └───────────────────────────────┘
                                  │
                                  ▼
                  ┌─────────────────────────────┐
                  │      SCHEDULE PPM           │
                  │  (Preventive Maintenance)   │
                  └─────────────────────────────┘
                                  │
                                  ▼
                  ┌─────────────────────────────┐
                  │      PPM PROCESSING         │
                  │  • Reset Accumulation Stroke │
                  │  • Record PPM History       │
                  │  • Update Die Status        │
                  └─────────────────────────────┘
```

---

3. Database Schema

3.1 Entity Relationship

```
┌──────────────┐       ┌──────────────┐       ┌──────────────┐
│  customers   │──1:N──│    dies      │──1:N──│production_   │
│              │       │              │       │   logs       │
└──────────────┘       └──────────────┘       └──────────────┘
                              │
                    ┌─────────┼─────────┐
                    │         │         │
                    ▼         ▼         ▼
             ┌──────────┐ ┌──────────┐ ┌──────────────┐
             │machines  │ │tonnage_  │ │ppm_schedules│
             │          │ │standards│ │              │
             └──────────┘ └──────────┘ └──────────────┘
                                              │
                                              ▼
                                       ┌──────────────┐
```

```
                                        │ppm_histories│
                                        └─────────────┘
```

3.2 Dies Table Structure

Column	Type	Description
`id`	BIGINT	Primary key
`part_number`	VARCHAR	Unique part identifier
`part_name`	VARCHAR	Part description
`customer_id`	BIGINT	FK to customers
`machine_id`	BIGINT	FK to machines
`accumulation_stroke`	INT	Current stroke count
`ppm_standard`	INT	Custom standard stroke (nullable)
`control_stroke`	INT	Override stroke limit (nullable)
`lot_size`	INT	Strokes per lot (default: 2500)
`ppm_alert_status`	ENUM	'none', 'orange', 'red'
`status`	VARCHAR	Die status

3.3 Key Fields Priority Chain

```
Standard Stroke Resolution:

    ┌──────────────────────────────────────────────────┐
    │  1. control_stroke (if set)                      │
    │     ↓ (if null)                                  │
    │  2. ppm_standard (if set)                        │
    │     ↓ (if null)                                  │
    │  3. tonnageStandard->standard_stroke (from machine) │
    │     ↓ (if null)                                  │
    │  4. Default: 10,000                              │
    └──────────────────────────────────────────────────┘
```

---

4. Core Logic

4.1 Standard Stroke Calculation

Location: `app/Models/DieModel.php`

```php
public function getStandardStrokeAttribute()
{
    // Priority 1: Control stroke (highest priority)
    if ($this->control_stroke) {
        return $this->control_stroke;
    }

    // Priority 2: PPM Standard (die-specific)
    if ($this->ppm_standard) {
        return $this->ppm_standard;
    }

    // Priority 3: Tonnage standard based on machine
    if ($this->machine && $this->machine->tonnageStandard) {
        return $this->machine->tonnageStandard->standard_stroke;
    }

    // Priority 4: Default value
    return 10000;
}
```

4.2 Lot Size Resolution

Location: `app/Models/DieModel.php`

```php
public function getLotSizeValueAttribute()
{
    // Priority 1: Die-specific lot_size
    if ($this->lot_size) {
        return $this->lot_size;
    }

    // Priority 2: Tonnage standard lot_size
    if ($this->machine && $this->machine->tonnageStandard) {
        return $this->machine->tonnageStandard->lot_size ?? 2500;
    }

    // Priority 3: Default
    return 2500;
}
```

4.3 Total Lots Calculation

```php
public function getTotalLotsAttribute()
{
    $lotSize = $this->lot_size_value;
    if ($lotSize <= 0) return 1;
```

```
    // IMPORTANT: Cast to int to avoid float comparison issues
    return (int) ceil($this->standard_stroke / $lotSize);
}
```

Example:

```
Standard Stroke: 10,000
Lot Size: 2,500
Total Lots: ceil(10,000 / 2,500) = 4 lots
```

4.4 PPM Status Determination

Location: `app/Models/DieModel.php`

```
public function getPpmStatusAttribute()
{
    $current = $this->accumulation_stroke ?? 0;
    $standard = $this->standard_stroke;
    $lotSize = $this->lot_size_value;

    // Calculate thresholds
    $orangeThreshold = $standard - $lotSize;  // e.g., 10,000 - 2,500 = 7,500
    $redThreshold = $standard;                 // e.g., 10,000

    if ($current >= $redThreshold) {
        return 'critical';   // 🔴 RED - Must stop production
    } elseif ($current >= $orangeThreshold) {
        return 'warning';    // 🟠 ORANGE - Schedule PPM soon
    }

    return 'normal';         // 🟢 GREEN - OK to continue
}
```

Visual Representation:

```
    0                    7,500              10,000
    |────────────────────|──────────────────|
    |     🟢 GREEN       |    🟠 ORANGE     |🔴 RED
    |     (Normal)       |   (Warning)      |(Critical)
    └────────────────────┴──────────────────┴──────────
                         ▲                  ▲
                  Orange Threshold    Red Threshold
                  (Standard - 1 Lot)  (Standard Stroke)
```

---

5. Alert System

5.1 Alert Flow

```
┌──────────────────────────────────────────────────────────┐
│                      ALERT TRIGGER                       │
│  When: Production log saved (stroke accumulation updated)│
└──────────────────────────────────────────────────────────┘
                              │
                              ▼
                ┌──────────────────────────┐
                │  Check accumulation_stroke │
                │  vs standard_stroke        │
                └──────────────────────────┘
                              │
                ┌─────────────┼─────────────┐
                │             │             │
                ▼             ▼             ▼
        ┌──────────┐   ┌──────────┐   ┌──────────┐
        │  < Orange │   │  Orange   │   │  >= Red   │
        │ Threshold │   │ Threshold │   │ Threshold │
        └──────────┘   └──────────┘   └──────────┘
              │             │             │
              ▼             ▼             ▼
        ┌──────────┐   ┌──────────┐   ┌──────────┐
        │  No Alert │   │ 🟠 Orange│   │ 🔴 Red   │
        │           │   │  Alert   │   │  Alert   │
        └──────────┘   └──────────┘   └──────────┘
                              │             │
                              ▼             ▼
                       ┌──────────┐   ┌──────────┐
                       │  Notify:  │   │  Notify:  │
                       │  • MGR/GM │   │  • MGR/GM │
                       │  • MD     │   │  • MD     │
                       │           │   │  • MTN    │
                       └──────────┘   └──────────┘
```

5.2 Alert Recipients

Location: `app/Services/AlertService.php`

```php
// Orange Alert Recipients
protected function getOrangeAlertRecipients()
{
    return User::where(function($query) {
        $query->where('role', User::ROLE_MGR_GM)
              ->orWhere('role', User::ROLE_MD);
    })->get();
}

// Red Alert Recipients (includes MTN Dies)
protected function getRedAlertRecipients()
```

```
{
    return User::where(function($query) {
        $query->where('role', User::ROLE_MGR_GM)
              ->orWhere('role', User::ROLE_MD)
              ->orWhere('role', User::ROLE_MTN_DIES);
    })->get();
}
```

5.3 Notification Content

Orange Alert Email:

```
Subject: ⚠️ PPM Warning Alert - {part_number}

Die {part_number} ({part_name}) is approaching PPM threshold.

Current Stroke: 8,000
Standard Stroke: 10,000
Usage: 80%

Action Required: Schedule PPM within next production cycle.
```

Red Alert Email:

```
Subject: 🚨 CRITICAL PPM Alert - {part_number}

Die {part_number} ({part_name}) has exceeded PPM threshold!

Current Stroke: 10,500
Standard Stroke: 10,000
Usage: 105%

IMMEDIATE ACTION REQUIRED: Stop production and perform PPM!
```

---

6. Lot Progress Visualization

6.1 Zone-Based Coloring Logic

Location: `app/Models/DieModel.php`

```
public function getLotProgressAttribute()
{
    $totalLots = $this->total_lots;      // e.g., 4
    $lotSize = $this->lot_size_value;    // e.g., 2,500
    $current = $this->accumulation_stroke ?? 0;
    $standard = $this->standard_stroke;  // e.g., 10,000
```

```
    $lots = [];

    for ($i = 1; $i <= $totalLots; $i++) {
        $lotStart = ($i - 1) * $lotSize;
        $lotEnd = $i * $lotSize;

        // Determine zone color based on position
        if ($i === $totalLots) {
            $zone = 'red';           // Last lot = RED
        } elseif ($i === $totalLots - 1) {
            $zone = 'orange';        // Second-to-last = ORANGE
        } else {
            $zone = 'green';         // Others = GREEN
        }

        // Determine if lot is filled based on current stroke
        $filled = $current >= $lotEnd;
        $partial = !$filled && $current > $lotStart;

        $lots[] = [
            'lot' => $i,
            'start' => $lotStart,
            'end' => $lotEnd,
            'zone' => $zone,
            'filled' => $filled,
            'partial' => $partial,
            'percentage' => $partial
                ? (($current - $lotStart) / $lotSize) * 100
                : ($filled ? 100 : 0),
        ];
    }

    return $lots;
}
```

6.2 Visual Example

Scenario: Standard = 10,000, Lot Size = 2,500, Current = 6,000

```
  Lot 1       Lot 2       Lot 3       Lot 4
(0-2,500)  (2,500-5,000) (5,000-7,500) (7,500-10,000)

┌──────────┐ ┌──────────┐ ┌──────────┐ ┌──────────┐
│▓▓▓▓▓▓▓▓▓▓│ │▓▓▓▓▓▓▓▓▓▓│ │▓▓▓▓▓▓▓▓▓▓│ │          │
│ ◍ 100%   │ │ ◍ 100%   │ │ ◍ 40%    │ │ ◍ 0%     │
└──────────┘ └──────────┘ └──────────┘ └──────────┘
    ✓            ✓          Partial      Empty
```

6.3 Frontend Component

Location: `resources/js/Components/PPM/LotProgress.jsx`

```
// Zone-based color classes
const getZoneColor = (zone, filled, partial) => {
    if (!filled && !partial) {
        // Empty lot - lighter version with border
        return {
            green: 'bg-green-100 border border-green-300',
            orange: 'bg-orange-100 border border-orange-300',
            red: 'bg-red-100 border border-red-300',
        }[zone];
    }

    // Filled lot - solid color
    return {
        green: 'bg-green-500',
        orange: 'bg-orange-500',
        red: 'bg-red-500',
    }[zone];
};
```

---

7. User Roles & Permissions

7.1 Role Definitions

Location: `app/Models/User.php`

```
const ROLE_ADMIN = 'admin';
const ROLE_PE = 'pe';              // Production Engineering
const ROLE_MTN_DIES = 'mtn_dies';  // Maintenance Dies
const ROLE_MGR_GM = 'mgr_gm';      // Manager / General Manager
const ROLE_MD = 'md';              // Managing Director
```

7.2 Role Permissions Matrix

Feature	Admin	PE	MTN Dies	MGR/GM	MD
View Dashboard	☑	☑	☑	☑	☑
Manage Dies	☑	✕	☑	✕	✕
Input Production Log	☑	☑	☑	✕	✕
Schedule PPM	☑	✕	☑	✕	✕
Process PPM	☑	✕	☑	✕	✕
Receive Orange Alert	✕	✕	✕	☑	☑

Feature	Admin	PE	MTN Dies	MGR/GM	MD
Receive Red Alert	❌	❌	☑	☑	☑
Manage Users	☑	❌	❌	❌	❌

7.3 Alert Recipients by Role

```
┌──────────────────────────────────────────────────────────┐
│                      ALERT RECIPIENTS                    │
├──────────────────────┬──────────────────┬────────────────┤
│       Role           │   Orange Alert   │    Red Alert   │
├──────────────────────┼──────────────────┼────────────────┤
│  Admin               │       ❌         │       ❌       │
│  PE                  │       ❌         │       ❌       │
│  MTN Dies            │       ❌         │       ☑        │
│  MGR/GM              │       ☑          │       ☑        │
│  MD                  │       ☑          │       ☑        │
└──────────────────────┴──────────────────┴────────────────┘
```

---

8. API Endpoints

8.1 Dies Management

Method	Endpoint	Description
GET	`/dies`	List all dies with pagination
GET	`/dies/{id}`	Show die details
POST	`/dies`	Create new die
PUT	`/dies/{id}`	Update die
DELETE	`/dies/{id}`	Delete die
POST	`/dies/{id}/schedule-ppm`	Schedule PPM for die
POST	`/dies/{id}/start-ppm`	Start PPM processing

8.2 Production Logs

Method	Endpoint	Description
GET	`/production`	List production logs
GET	`/production/{id}`	Show log details
POST	`/production`	Create new log (adds stroke)
PUT	`/production/{id}`	Update log (adjusts stroke)

Method	Endpoint	Description
DELETE	`/production/{id}`	Delete log (decrements stroke)

8.3 PPM Schedule

Method	Endpoint	Description
GET	`/ppm-schedule`	List PPM schedules
POST	`/ppm-schedule`	Create PPM schedule
PUT	`/ppm-schedule/{id}`	Update schedule
POST	`/ppm-schedule/{id}/complete`	Mark PPM complete

Appendix A: Production Log Stroke Adjustment

When editing or deleting production logs, the system automatically adjusts die accumulation:

On Update:

```
$difference = $newOutputQty - $oldOutputQty;
$die->accumulation_stroke += $difference;
```

On Delete:

```
$die->accumulation_stroke -= $productionLog->output_qty;
```

Appendix B: PPM Processing Flow

```
┌──────────────────────────────────────────────────────────┐
│                      PPM PROCESSING                      │
└──────────────────────────────────────────────────────────┘
                              │
                              │
                              ▼
                 ┌──────────────────────────────┐
                 │  1. Create PPM History       │
                 │     - Record current stroke  │
                 │     - Record maintenance     │
                 │       performed              │
                 └──────────────────────────────┘
                              │
                              │
                              ▼
                 ┌──────────────────────────────┐
                 │  2. Reset Die Accumulation   │
```

```
                  │     accumulation_stroke = 0   │
                  └──────────────────────────────┘
                                  │
                                  │
                                  ▼

                  ┌──────────────────────────────┐
                  │  3. Update Die Status         │
                  │     ppm_alert_status = none   │
                  │     status = 'active'         │
                  └──────────────────────────────┘
                                  │
                                  │
                                  ▼

                  ┌──────────────────────────────┐
                  │  4. Update PPM Schedule       │
                  │     status = 'completed'      │
                  │     completed_at = now()      │
                  └──────────────────────────────┘
```

Appendix C: Configuration Constants

```
// Default Values
const DEFAULT_STANDARD_STROKE = 10000;
const DEFAULT_LOT_SIZE = 2500;

// Alert Thresholds
const ORANGE_THRESHOLD = STANDARD_STROKE - LOT_SIZE;  // 75%
const RED_THRESHOLD = STANDARD_STROKE;                 // 100%

// Tonnage Standards (Machine-based)
// 800T  → Standard: 10,000, Lot: 2,500
// 1200T → Standard: 8,000,  Lot: 2,000
// 1600T → Standard: 6,000,  Lot: 1,500
```

Document End

For questions or clarifications, contact the Development Team.